

AUTOSAR High-Level Design (HLD)

Sample Test Document for AUTOSAR HLD Document Analysis Assistant

This document is a small synthetic AUTOSAR-style HLD created specifically for testing PDF ingestion, entity extraction, RAG question answering, dependency mapping, and inconsistency detection. It is not an official AUTOSAR specification.

1. Engine Control Architecture

EngineControlSWC is responsible for calculating fuel injection, spark timing, and target torque. It receives engine information from **SensorManager** and sends control information to **ControlManager**.

SensorManager collects engine sensor information. It provides the **SensorDataInterface** through the **SensorDataPPort**. The signals are **EngineSpeed**, **EngineTemperature**, and **ThrottlePosition**.

EngineControlSWC receives **SensorDataInterface** through its **SensorDataRPort**. It produces **TargetTorque** through the **EngineDataPPort**.

2. Control Architecture

ControlManager receives **TargetTorque** from **EngineControlSWC** and sends commands to **Actuator**. The main control flow is: **SensorManager** → **EngineControlSWC** → **ControlManager** → **Actuator**.

The **EngineDataInterface** carries **TargetTorque** from **EngineControlSWC** to **ControlManager**. **ControlManager** receives this interface through the **EngineDataRPort**.

3. Diagnostic Architecture

DiagnosticManager handles diagnostic requests and coordinates diagnostic services. **Dem** manages diagnostic events and fault information. **Dcm** manages diagnostic communication. **CanIf** provides the CAN interface to the communication path.

The diagnostic flow is: **DiagnosticManager** → **Dcm** → **CanIf**. **Dem** provides diagnostic event information to **DiagnosticManager**.

The diagnostic interface contains the signals **DtcStatus**, **DiagnosticSessionState**, and **FaultRecordData**.

4. Communication Dependencies

The main dependencies in the system are listed below.

Source	Relationship	Target
SensorManager	provides data to	EngineControlSWC
EngineControlSWC	sends TargetTorque to	ControlManager
ControlManager	sends control commands to	Actuator
DiagnosticManager	uses diagnostic services from	Dem
DiagnosticManager	sends diagnostic requests to	Dcm
Dcm	uses CAN communication through	CanIf

5. Interface and Port Summary

SensorDataInterface: provided by **SensorManager** through **SensorDataPPort** and required by **EngineControlSWC** through **SensorDataRPort**.

EngineDataInterface: provided by EngineControlSWC through EngineDataPPort and required by ControlManager through EngineDataRPort.

DiagnosticInterface: used by DiagnosticManager, Dcm, and Dem for diagnostic information.

6. Important Signals

Signal	Purpose
EngineSpeed	Current engine speed
EngineTemperature	Engine temperature value
ThrottlePosition	Throttle position value
TargetTorque	Requested engine torque
DtcStatus	Diagnostic trouble-code status
DiagnosticSessionState	Current diagnostic session
FaultRecordData	Stored diagnostic fault information

7. Architecture Review Note

During review, engineers should verify that every referenced component has a definition, that each interface has the correct provider and consumer, and that all dependencies are consistent across the HLD.